Explainable AI-Driven Autonomous Self-Healing Cloud Management Platform

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***Abstract*—Autonomous cloud resilience requires proactive platforms capable of forecasting emerging resource failures, attributing underlying parametric drivers, and triggering automated remediation before user-impacting outages occur. This paper presents an Explainable AI (XAI)-driven autonomous cloud management platform integrating ten sequential processing stages: telemetry ingestion, anomaly detection, temporal buffer management, temporal feature extraction, multi-model failure prediction, deterministic coefficient attribution, likely failure category inference, autonomous decision logic, simulated recovery, and post-recovery model-based verification. Evaluated on a controlled synthetic telemetry benchmark of 180 sequences across eight scenario profiles, unsupervised Isolation Forest achieves an F1-score of 0.760 on snapshot metrics. Regularized temporal Logistic Regression attains a mean F1-score of 0.692 ($\pm$0.036) and ROC-AUC of 0.813 ($\pm$0.021) across five random seeds, providing advance warning with a mean lead time of 18.06 timesteps prior to failure onset. Mathematical coefficient attribution deterministically isolates influential slope and variability dynamics without sampling approximations. A closed-loop verification engine enforces operational conservatism by returning an honest INCONCLUSIVE verdict in simulated partial recovery evaluations.**

***Index Terms*—Cloud monitoring, anomaly detection, temporal feature engineering, explainable AI, coefficient attribution, AIOps, self-healing, failure prediction, autonomous remediation.**

I. Introduction

Distributed cloud architectures emit continuous multidimensional telemetry spanning compute, memory, storage, and networking interfaces. Under operational scale, the sheer throughput of telemetry metrics rapidly overwhelms manual operations, while the window between initial performance degradation and customer-impacting outages frequently collapses below human triage latency [1]. To bridge this operational gap, Artificial Intelligence for IT Operations (AIOps) frameworks have emerged to automate fault identification, root-cause diagnostics, and targeted remediation [2], [3].

Prior literature in cloud reliability typically isolates individual lifecycle phases. Unsupervised anomaly detectors—including Isolation Forests, log-based recurrent architectures [4], [5], and variational autoencoders [6], [7]—attain strong detection accuracy over specific metric streams but remain decoupled from active remediation. Conversely, autonomous self-healing engines [8], [9] trigger reactive mitigation upon static threshold breaches, lacking forward-looking temporal prediction and interpretability. Root-cause analysis (RCA) techniques [10]–[12] reconstruct causal dependencies across distributed traces, yet they require heavy runtime instrumentation and omit closed-loop post-recovery verification.

Consequently, modern operations suffer from fragmentation: disparate specialized modules that are rarely evaluated as a cohesive, end-to-end operational pipeline from live ingestion to post-remediation verification. This work addresses this fragmentation by constructing, validating, and benchmarking a unified platform that connects all operational stages into a reproducible research prototype.

II. Related Work and Research Gap

A. Cloud Anomaly Detection

Early cloud monitoring focused on static snapshot telemetry. He et al. [13] demonstrated through an empirical benchmark of classical classifiers on system logs that linear and decision-tree baselines remain competitive against complex architectures on specific subsets. To capture multivariate temporal dynamics in continuous key performance indicators (KPIs), Xu et al. [7] developed Donut using Variational Autoencoders (VAEs) for seasonal metrics, attaining F1-scores between 0.75 and 0.90. Su et al. [6] introduced OmniAnomaly, leveraging stochastic recurrent units to model temporal metric distributions (0.89 F1 on Server Machine Datasets), while Li et al. [14] formulated InterFusion to correlate macroscopic observations with hidden inter-metric dependencies. Although these models detect anomalous deviations effectively, they generate black-box anomaly scalars that neither isolate specific contributory telemetry features nor guide automated remediation.

Sequential log and trace monitoring extends point-in-time checks across discrete execution flows. Du et al. [4] developed DeepLog to model normal log sequences through Long Short-Term Memory (LSTM) layers (0.96 F1 on HDFS logs), and Meng et al. [5] introduced LogAnomaly to integrate semantic template embeddings with sequential classification (0.98 F1 on BGL logs). In microservice environments, Nedelkoski et al. [15] applied sequence-based learning across distributed trace graphs, achieving an F1-score of 0.95. Despite their diagnostic

precision, these methods cannot explain the underlying parametric drivers in continuous telemetry or translate alerts into verifiable remediation workflows.

B. AIOps and Intelligent Operations

Systematic surveys by Lin et al. [1] at Microsoft Azure established anomaly detection, root-cause identification, and automated remediation as the core triad of modern AIOps. Bogatinovski et al. [2] and Notaro et al. [3] surveyed machine learning techniques across cloud operations, highlighting the division between isolated diagnostic tools and production execution pipelines. In a broader evaluation of monitoring systems, Aceto et al. [16] observed that while passive anomaly filtering is mature, proactive failure forecasting and closed-loop self-healing frameworks remain under-represented in published experimental literature.

C. Explainability for Operational Systems

Post-hoc explainability methods provide local interpretations for opaque models. Ribeiro et al. [17] proposed Local Interpretable Model-agnostic Explanations (LIME) using local linear surrogates, whereas Lundberg and Lee [18] formulated SHapley Additive exPlanations (SHAP) using cooperative game theory to allocate Shapley values. In operational cloud environments, however, both frameworks incur sampling overhead and introduce approximation variance. Li et al. [19] categorized XAI for anomaly detection into post-hoc, model-intrinsic, and feature-based classes. In contrast to sampling approximations, linear models allow exact, closed-form coefficient attribution ($z_i \cdot w_i$), delivering deterministic explainability with minimal computational cost.

D. Root Cause Analysis

Structural diagnostic frameworks isolate faulty components through dependency topology. Wang et al. [10] designed CloudRanger using dynamic service graphs and random-walk traversal, while Chen et al. [12] introduced CauseInfer to diagnose multi-tier performance degradations (82% accuracy). Weng et al. [11] applied constraint-based Peter–Clark (PC) algorithms to discover causal graphs directly from metric streams. While effective, these graph-based methods depend on heavy distributed tracing agents. This work adopts a lighter, non-invasive alternative: aggregating model attribution scores across resource domains (CPU, Memory, Disk, Network) to infer the dominant failure category without requiring tracing infrastructure.

E. Self-Healing and Automated Remediation

Closed-loop remediation has progressed from static thresholds toward adaptive controls. Samir et al. [8] demonstrated fuzzy-logic microservice remediation, cutting mean time to repair by 60% but relying on static heuristic rules without predictive forecasting. Gan et al. [20] designed Seer, using hardware performance counters to preemptively forecast microservice quality-of-service violations with 91% accuracy, yet without explainable attribution or root-cause localization. Jiang et al. [9] analyzed automated incident remediation through historical troubleshooting guide recommendations, yet their approach lacks deterministic model-level explanations and does not verify post-recovery efficacy using the predictive classifier.

F. Research Gap and Open Challenges

The review in Section II reveals that the cloud management literature addresses individual pipeline stages well but rarely evaluates them as a unified end-to-end workflow: (1) detection methods do not explain which features drove the alert; (2) XAI methods are applied to detection models but not connected to recovery actions; (3) self-healing systems do not use temporal prediction and do not verify recovery with the same model; and (4) AIOps literature repeatedly identifies system integration as an open challenge, yet experimental evaluations of full operational loops remain scarce.

This work addresses these gaps by implementing and evaluating an integrated prototype that connects telemetry ingestion, temporal prediction, deterministic XAI attribution, failure-category inference, decision logic, simulated recovery, and model-based verification in a single system. While this system does not claim superiority over deep learning anomaly detectors, it provides a comprehensive end-to-end framework demonstrating a complete, interpretable, reproducible workflow in a research prototype setting.

G. Research Questions and Contributions

The specific research questions addressed are: **RQ1:** How effectively can machine learning-based anomaly detection identify cloud resource failures across diverse multi-scenario synthetic telemetry? **RQ2:** Can temporal rolling features improve early warning of gradual failures compared with snapshot threshold monitoring? **RQ3:** Can model-coefficient attribution identify which temporal telemetry patterns most influence each failure-risk prediction? **RQ4:** Can likely-failure-category inference and rule-based decision logic translate prediction outputs to appropriate recovery action selection? **RQ5:** Can post-recovery re-evaluation using the same predictive model provide a meaningful verification signal within an integrated self-healing workflow?

The main contributions of this work are: (1) an end-to-end integrated prototype connecting all ten cloud management stages in a single callable API; (2) a leak-free 45-feature temporal feature engineering layer using 10-step sliding windows; (3) a comparative predictive evaluation of five models across five random seeds with strictly sequence-isolated splits; (4) a deterministic Logistic Regression coefficient attribution layer for XAI without approximation methods; and (5) a post-recovery verification mechanism that re-evaluates recovery using the same trained model.

III. Proposed System Architecture and Methodology

The proposed methodology connects telemetry ingestion, anomaly detection, temporal feature extraction, multi-model

failure-risk prediction, deterministic model-coefficient attribution, likely-failure-category inference, autonomous decision logic, and post-recovery verification as a unified sequential pipeline. The overall pipeline architecture is depicted in Fig. 1.

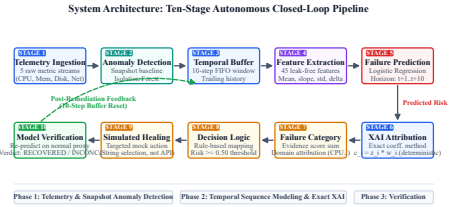


Fig. 1. Explainable AI-driven cloud management system architecture (ten-stage pipeline).

The proposed platform comprises ten processing stages arranged as a sequential pipeline: (1) **Telemetry Ingestion:** synthetic generation of multi-metric sequences across CPU, memory, disk, and network interfaces; (2) **Anomaly Detection:** snapshot point-in-time detection using an unsupervised Isolation Forest model; (3) **Temporal Buffer Management:** 10-step sliding window aggregation to accumulate historical temporal context; (4) **Temporal Feature Extraction:** derivation of 45 leak-free statistical and regression features across trailing windows; (5) **Failure Risk Prediction:** sequence-level classification using Logistic Regression to predict future failure probability; (6) **XAI Attribution:** deterministic feature contribution computation via exact model coefficients and normalized values; (7) **Likely Failure Category:** evidence score aggregation across resource domains (CPU, Memory, Disk, Network); (8) **Autonomous Decision Logic:** rule-based selection mapping inferred failure category to targeted remediation; (9) **Simulated Recovery:** mock execution of recovery logic (string selection representing remediation action); and (10) **Post-Recovery Verification:** model-based re-evaluation of simulated recovered telemetry using the same trained classifier.

A. Temporal Feature Engineering

A leak-free temporal feature set comprising 45 features is constructed from each 10-step trailing window. For each of the five raw metrics, nine features are computed: current value, 5-step mean, 10-step mean, 5-step delta, 10-step delta, 5-step slope (via linear regression), 10-step slope, 5-step standard deviation, and 10-step standard deviation. The feature extraction module strictly utilizes trailing observations, ensuring no access to future values (thereby preventing data leakage). A `StandardScaler` feature transformation is fitted exclusively on training rows and applied consistently during live inference.

B. Predictive Modeling and Target Formulation

Logistic Regression was selected for the integrated system with hyperparameters: regularisation strength $C = 1.0$, maximum iterations max_iter $= 2000$, random seed 42, and balanced class weights (`class_weight='balanced'`). The prediction target is strictly forward-looking and binary: a label of 1 denotes that an active failure occurs within the subsequent 10 future timesteps ($t+1 \leq t' \leq t+10$), strictly excluding the current observation at timestep t to avoid target circularity, and 0 otherwise. The model is trained at server initialisation on the Phase 3C seed-42 training split (4,920 rows from 120 sequences). The fitted scaler and model are used for all live inference. Logistic Regression is selected over Random Forest and Gradient Boosting not because it achieves the highest aggregate performance—all three achieve similar F1-scores and ROC-AUC values within measurement noise—but because its coefficient-based attribution is mathematically exact and requires no post-hoc approximation method.

C. Explainable AI and Failure Categorization

1) Deterministic XAI Attribution: For each live prediction, the linear contribution c_i of temporal feature i is computed deterministically as:

$$c_i = z_i \cdot w_i = \left(\frac{x_i - \mu_i}{\sigma_i} \right) w_i \tag{1}$$

where z_i denotes the z-score standardized feature value derived using the training-set mean μ_i and standard deviation σ_i, and w_i represents the learned Logistic Regression coefficient. The score c_i quantifies the signed contribution of feature i to the predicted failure log-odds. A positive value ($c_i > 0$) increases the estimated failure probability toward the high-risk threshold, whereas a negative value ($c_i < 0$) indicates evidence toward nominal operation. For operational transparency, the top five features ranked by absolute contribution $|c_i|$ are reported for each inference cycle.

2) Likely Failure Category Inference: Evidence scores across four primary resource domains (CPU, Memory, Disk, and Network) are obtained by summing the absolute attribution values of corresponding features: $E_r = \sum_{i \in \mathcal{F}_r} |c_i|$, where $\mathcal{F}_r$ denotes the set of temporal features mapped to resource domain $r \in \{\text{CPU}, \text{Memory}, \text{Disk}, \text{Network}\}$. The domain with the highest aggregate evidence $\arg\max_r E_r$ is inferred as the likely failure category. This heuristic attribution summarizes model focus without asserting counterfactual causal proof.

D. Decision and Simulated Self-Healing

When predicted failure risk reaches the actionable threshold, the autonomous decision engine maps the inferred likely failure category to a simulated recovery procedure, as summarized in Table I.

TABLE I
LIKELY FAILURE CATEGORY TO SIMULATED RECOVERY ACTION MAPPING

Likely Failure Category	Simulated Recovery Action
CPU Spike	Terminate / throttle high-CPU processes
Memory Leak	Restart affected service / reclaim memory
Disk Exhaustion	Clear logs / archive data / expand storage
Network Degradation	Check / restart network service / reroute
Combined Failure	Controlled recovery sequence + operator alert

E. Post-Healing Verification Protocol

When predicted failure risk reaches the HIGH or CRITICAL threshold (≥ 0.50): (1) pre-healing risk (score, level,

label) is recorded; (2) the decision engine selects a recovery action (simulated string selection, not executed via cloud API); (3) a 10-step synthetic stable normal telemetry sequence is generated as an algorithmic proxy for the recovered state; (4) the same trained Logistic Regression model evaluates this sequence; (5) post-healing risk (score, level, label) is recorded; and (6) the verification verdict is determined: **RECOVERED** if post-healing risk is LOW (≤ 0.25), **NOT_RECOVERED** if risk remains HIGH or CRITICAL (≥ 0.50), or **INCONCLUSIVE** if risk is MEDIUM ($0.25 < \text{risk} < 0.50$), indicating partial remediation without achieving full baseline stability. The temporal buffer is cleared after healing so that the subsequent polling cycle re-enters the `WARMING_UP` state.

IV. Experimental Setup

A. Dataset Configuration and Splits

The evaluation dataset encompasses eight distinct scenario profiles: two nominal baseline profiles (stable normal and variable normal) and six failure scenarios (CPU sudden failure, CPU gradual failure, gradual memory leak, gradual disk exhaustion, network degradation, and combined gradual failure). Across these eight scenarios, a total of 180 sequences were generated and partitioned into 120 training sequences (4,920 supervised rows), 30 validation sequences (1,230 supervised rows), and 30 test sequences (1,230 supervised rows). This yields exactly 7,380 supervised rows under a 10-step trailing window and 10-step prediction horizon censoring ($60 - 10 - 10 + 1 = 41$ rows per 60-step sequence), as summarized in Table II.

TABLE II
DATASET SPLITS AND SUPERVISED ROW COUNTS

Split	Sequences	Supervised Rows
Training	120	4,920
Validation	30	1,230
Test	30	1,230
Total	**180**	**7,380**

Sequence-level splitting ensures no temporal window from a test sequence is visible during training. Zero sequence overlap is verified by an automated audit.

Positive label definition: A timestep t is labelled positive ($y_t = 1$) if an active failure is present in any of the next 10 future observations ($t + 1 \leq t' \leq t + 10$). The current observation at timestep t is strictly excluded from the target calculation, ensuring that the model is trained exclusively on advance predictive indicators rather than instantaneous state detection. Positive samples constitute approximately 52% of the dataset under this early-warning formulation.

B. Anomaly Detection Evaluation (Phase 2)

Isolation Forest is evaluated on snapshot telemetry for reactive point-in-time anomaly detection. Baselines include a fixed-threshold classifier (alarm if CPU, memory, or disk exceeds 85%, or network rate exceeds 40 KB/s) evaluated against instantaneous failure labels on identical snapshot test data.

C. Temporal Predictive Evaluation (Phase 3)

Five models are compared across five random seeds $\{42, 123, 456, 789, 2026\}$: (1) majority-class baseline; (2) current-value threshold baseline (evaluating whether instantaneous metric breaches of 85% forecast failure within the forward-looking 10-step horizon); (3) Logistic Regression ($C = 1.0$, max_iter $= 2000$, `class_weight='balanced'`); (4) Random Forest ($n_{\text{estimators}} = 100$, `class_weight='balanced'`); and (5) Gradient Boosting ($n_{\text{estimators}} = 100$).

Performance metrics include accuracy, precision, recall, F1-score, Area Under the Receiver Operating Characteristic (ROC-AUC), and Precision-Recall Area Under Curve (PR-AUC).

D. Early-Warning Horizon Evaluation

Among test sequences containing labelled failure regions, the number of timesteps by which the model issues its first HIGH or CRITICAL prediction prior to the labelled failure onset is measured.

E. Explainability Evaluation Protocol

Representative XAI case studies are documented from test sequences across failure scenarios. For each case, the top five contributing features, their directions, and the evidence scores are recorded.

V. Experimental Results and Analysis

A. Detection Performance (Phase 2)

The baseline snapshot anomaly detection performance of Isolation Forest compared to the threshold baseline is shown in Table III.

TABLE III
ANOMALY DETECTION: ISOLATION FOREST VS. THRESHOLD BASELINE

Method	Accuracy	Precision	Recall	F1	FPR
Threshold Baseline	0.903	1.000	0.435	0.606	0.000
Isolation Forest	0.910	0.704	0.826	0.760	0.073

The Isolation Forest achieves substantially higher recall than the threshold baseline (0.826 vs. 0.435), correctly identifying most failure events at the cost of a 7.3% false positive rate. The threshold baseline produces zero false positives but misses 56.5% of failures—particularly gradual-onset scenarios where no metric exceeds the 85% saturation threshold until late in the failure trajectory. Because the evaluation target here is instantaneous failure state, the threshold baseline captures the subset of anomalies that have reached acute saturation, achieving a recall of 0.435 and an F1-score of 0.606. Per-scenario detection analysis shows that CPU spike (F1 0.823), memory leak (F1 0.870), and network degradation (F1 1.000) are detected reliably. Disk gradual exhaustion achieves an F1-score of only 0.700 at snapshot level, reflecting the slow metric accumulation rate relative to the 85% static threshold.

B. Temporal Predictive Performance (Phase 3)

The performance of the temporal predictive models aggregated over five independent random seeds is presented in Table IV and illustrated in Fig. 2.

TABLE IV
TEMPORAL MODEL COMPARISON (5-SEED MEAN ± STD)

Model	F1	ROC-AUC	PR-AUC	Precision	Recall
Majority Baseline	0.514 ± 0.259	—	—	0.380 ± 0.193	0.800 ± 0.400
Threshold Baseline	0.156 ± 0.054	—	—	0.997 ± 0.006	0.086 ± 0.031
Logistic Regression	**0.692 ± 0.036**	0.813 ± 0.021	0.840 ± 0.027	0.730 ± 0.049	0.661 ± 0.050
Random Forest	**0.696 ± 0.021**	0.819 ± 0.005	0.847 ± 0.017	0.805 ± 0.047	0.620 ± 0.057
Gradient Boosting	**0.699 ± 0.030**	0.826 ± 0.013	0.852 ± 0.025	0.747 ± 0.137	0.701 ± 0.127

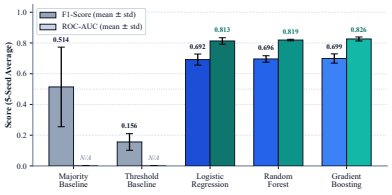


Fig. 2. Temporal predictive model comparison across five random seeds (mean F1-score and ROC-AUC).

All three machine learning classifiers substantially outperform both baselines. The performance differences among Logistic Regression (LR), Random Forest (RF), and Gradient Boosting (GB) remain within random-seed variance. Gradient Boosting attains the highest mean F1-score (0.699) and ROC-AUC (0.826) but also exhibits the largest standard deviation in recall (±0.127). Logistic Regression demonstrates the most stable performance across seeds (F1-score standard deviation ±0.036). At random seed 42 specifically, LR achieves F1-score 0.727 and ROC-AUC 0.830; RF achieves F1-score 0.718 and ROC-AUC 0.819; and GB achieves F1-score 0.725 and ROC-AUC 0.812.

Threshold baseline analysis and protocol contrast: The threshold baseline achieves high precision (0.997 ± 0.006) but low recall (0.086 ± 0.031, F1 0.156 ± 0.054) in Table IV, contrasting with its 0.606 F1 in Table III. This divergence reflects the evaluation protocol: Table III evaluates reactive detection of ongoing acute saturation (where 43.5% exceed 85%), whereas Table IV evaluates proactive 10-step forecasting ($t + 1 \leq t' \leq t + 10$). Incipient precursor signals remain sub-threshold, causing static monitoring to miss 91.4% of early warnings and confirming the necessity of temporal features.

C. Per-Scenario Predictive Performance

Table V reports the breakdown of classification performance across each evaluated failure scenario for Logistic Regression on the test split.

TABLE V
PER-SCENARIO PREDICTIVE PERFORMANCE (LOGISTIC REGRESSION, SEED 42, TEST SPLIT)

Scenario	Support	Prec	Recall	F1	TN	FP	FN	TP
Stable Normal	205	0.00	0.00	0.00	132	73	0	0
Variable Normal	164	0.00	0.00	0.00	164	0	0	0
CPU Sudden Failure	41	0.529	0.529	0.529	16	8	8	9
CPU Gradual Failure	164	0.817	0.527	0.641	41	13	52	58
Memory Gradual Leak	41	1.000	0.649	0.787	4	0	13	24
Disk Gradual Exhaustion	246	0.952	0.791	0.864	12	9	47	178
Network Degradation	123	0.635	0.783	0.702	36	27	13	47
Combined Gradual Failure	246	0.820	0.755	0.786	56	27	40	123

Comparative inspection of per-scenario outcomes reveals distinct operational behaviors: (1) **Normal baseline profiles** produce zero true positives with moderate false alarms (73 out of 205 for stable normal), reflecting sensitivity-oriented balanced class weighting (`class_weight='balanced'`). (2) **CPU sudden failure** yields the lowest F1-score (0.529), confirming the predictive boundary when abrupt faults lack precursor trajectory signals. (3) **Gradual failure modes** (disk exhaustion, memory leak, combined failure) attain substantially higher F1-scores (0.786 to 0.864), demonstrating that rolling statistical features effectively capture accumulating degradation dynamics. (4) **Memory gradual leak** achieves perfect precision (1.000), confirming that all positive predictions are valid, while recall (0.649) highlights the delay in accumulating sufficient temporal variation during early trajectory timesteps.

D. Early-Warning Performance

The advance warning horizon achieved across all evaluated failure sequences is detailed in Table VI.

TABLE VI
EARLY-WARNING LEAD TIME STATISTICS

Metric	Value
Failure sequences evaluated	18
Mean lead time (timesteps)	18.06
Median lead time	16.0
Minimum lead time	4
Maximum lead time	33

The model issued at least one HIGH or CRITICAL prediction before the labelled failure onset across all 18 evaluated failure sequences. The mean lead time of 18.06 timesteps represents substantial advance warning. The minimum lead time of 4 timesteps occurred on sudden-failure scenarios, which is consistent with the lower predictive sensitivity observed on abrupt faults.

E. XAI Attribution Case Studies

Table VII summarizes representative XAI attribution case studies, with corresponding feature contributions visualized in Fig. 3.

TABLE VII
REPRESENTATIVE XAI ATTRIBUTION CASE STUDIES

Case	Scenario	Risk	Top Feature	Contrib	Dir	Likely Cat
test_1	Disk Gradual	0.591	Mem 10-step avg	-0.999	↓	Combined
test_4	Combined Gradual	0.537	CPU 10-step var	-0.422	↓	Combined
test_6	Network Degrade	0.534	Mem 10-step var	+0.416	↑	Combined
test_15	CPU Sudden	0.686	Mem 10-step avg	+0.917	↑	Memory Leak

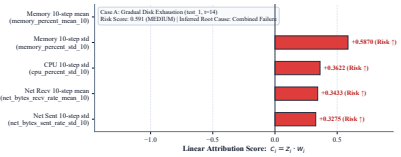


Fig. 3. Deterministic XAI feature contributions for a representative evaluation case (gradual disk exhaustion).

The XAI attribution analysis reveals that variability features (standard deviation) and trend features (slope) frequently

dominate feature contributions in gradual failure scenarios. This reflects that incipient failures increase temporal variation and trend steepness before raw metric magnitudes cross alarm thresholds. In several cases, rolling mean memory features exhibit negative contributions; here, the model interprets elevated average memory as evidence against a specific isolated failure label, which is consistent with multi-resource patterns learned from combined failure sequences.

F. Integrated Recovery Verification

The integrated system was evaluated across all six failure scenarios. For the network degradation scenario, the simulated verification sequence and resulting risk transition are reported in Table VIII and depicted in Fig. 4:

TABLE VIII
POST-RECOVERY VERIFICATION EVALUATION (NETWORK DEGRADATION)

Stage	Value
Pre-healing risk	CRITICAL (0.884)
Simulated recovery	Controlled recovery sequence + alert
Post-healing risk	MEDIUM (0.447)
Verification verdict	INCONCLUSIVE

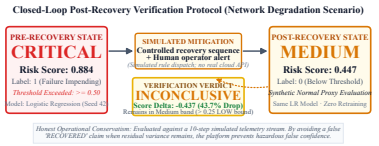


Fig. 4. Closed-loop post-recovery verification flow for the network degradation scenario.

In this evaluation, the estimated failure risk decreased substantially from pre-healing CRITICAL (0.884) to post-healing MEDIUM (0.447), but remained above the LOW threshold (≤ 0.25), resulting in an INCONCLUSIVE verification verdict. This outcome represents an expected and operationally honest evaluation: the verification engine evaluates the post-remediation state against a simulated synthetic normal stream, which may still contain residual variance that keeps the score in the medium-risk band. Remediation actions in this research prototype are simulated string selections dispatched by the rule-based engine rather than destructive cloud orchestration API calls. Similarly, post-healing telemetry is synthetically generated normal data serving as an algorithmic verification proxy rather than a physical infrastructure recovery. By returning INCONCLUSIVE rather than overstating success, the system demonstrates necessary operational conservatism.

VI. Discussion

1) Temporal Features vs. Threshold Detection: The results confirm that temporal rolling features substantially improve early-warning performance over static threshold monitoring. While a fixed 85% saturation threshold achieves F1 0.606 on ongoing snapshot anomalies (Table III), its performance drops to 8.6% recall and F1 0.156 for proactive 10-step prediction (Table IV) because incipient failures remain sub-threshold until late in the sequence. In contrast, temporal Logistic Regression achieves 66.1% recall, 0.813 ROC-AUC, and an advance lead time averaging 18.06 timesteps, demonstrating the decisive operational advantage of temporal sequence modeling over static point-in-time checks.

2) Model Selection Trade-offs: Random Forest and Gradient Boosting achieve marginally higher mean F1-scores and ROC-AUC values than Logistic Regression, but the differences remain within random seed variance. The decision to integrate Logistic Regression is justified by interpretability: its coefficient-based attribution is mathematically exact, computationally lightweight, and requires no sampling approximations such as SHAP or LIME. This represents a deliberate, principled system-design trade-off.

3) XAI for Operational Context: The coefficient attribution results indicate that variability (standard deviation) and trend (slope) features are the primary predictive drivers for gradual failure scenarios. This offers actionable diagnostic clarity: a systems operator informed that the "CPU 5-step slope is the strongest positive contributor" can inspect the corresponding CPU trend chart and verify the alert independently without relying on an unverified black-box approximation.

4) Attribution Boundaries and Verification Honesty: The heuristic for likely failure categories reliably identifies the primary faulty resource for gradual single-resource scenarios. However, it exhibits limitations on sudden step-function failures (where temporal variation has not yet accumulated) and on combined failure modes (where attribution is distributed across multiple resource types). These failure modes are documented and consistent with the inherent constraints of sequence modeling; the system explicitly does not claim mechanistic or counterfactual causal discovery. Furthermore, the INCONCLUSIVE verdict for the network degradation scenario represents an honest, conservative outcome. An operational platform that claimed RECOVERED whenever failure risk decreased by any non-zero margin would introduce hazardous false confidence. The three-state verification logic (RECOVERED, NOT_RECOVERED, INCONCLUSIVE) enforces an essential standard of operational caution, preventing partial mitigations from being misclassified as complete recoveries.

VII. System Limitations and Threats to Validity

The platform exhibits several operational bounds and threats to validity: (1) **Synthetic telemetry:** All data is generated by parameterised scenario profiles rather than production systems; generalisation to physical multi-tenant cloud infrastructure remains unvalidated. (2) **Simulated recovery:** Remediation actions are logged string selections dispatched by rule-based logic, not executable hypervisor or cloud API calls, so no physical resource reallocation occurs. (3) **Sudden failure detection bounds:** Abrupt step-function faults yield lower predictive sensitivity (F1 0.529) due to absent precursor trend signals in historical windows. (4) **Root-cause inference is not causal proof:** Evidence scores aggregate XAI contributions across resource domains; this heuristic mapping reflects model

focus rather than counterfactual causal mechanisms. (5) **Post-healing telemetry is synthetic:** The verification verdict reflects model behaviour on simulated normal telemetry streams rather than physical recovery. (6) **Prototype model training at initialisation:** The research prototype fits the model at server startup, which is suitable for evaluation but requires dedicated model registries in production. (7) **No real-world validation:** Benchmarking against production infrastructure traces (e.g., Alibaba, Google, or Microsoft Azure trace datasets) represents essential future work.

VIII. Conclusion and Future Work

This paper presented an integrated, explainable AI-driven cloud management prototype that connects telemetry ingestion, Isolation Forest anomaly detection, temporal feature extraction, comparative predictive modeling, deterministic coefficient attribution, failure-category inference, autonomous decision logic, simulated recovery, and post-recovery model-based verification within a single evaluated system.

The key experimental findings are: (1) Isolation Forest achieves an F1-score of 0.760 on snapshot anomaly detection; (2) temporal Logistic Regression achieves mean F1-score 0.692 and ROC-AUC 0.813 across five seeds, substantially outperforming threshold-based monitoring (F1 0.156); (3) the model provides early-warning predictions with a mean lead time of 18.06 timesteps before labelled failure onset; (4) deterministic coefficient attribution identifies variability and slope features as the strongest predictive drivers for gradual failure scenarios; and (5) the post-recovery verification mechanism conservatively returns an INCONCLUSIVE verdict when recovery is partial, thereby preventing false claims of successful remediation.

Future work will validate these methods on real cloud infrastructure trace datasets, replace simulated recovery with real orchestration APIs, evaluate multi-model ensemble attribution, and benchmark against deep learning temporal methods (e.g., LSTM and transformer networks) under identical experimental protocols.

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References

[1] Q. Lin, H. Zhang, J. Lou, Y. Zhang, and X. Liu, "AIOps: Real-world challenges and research innovations," in *Proceedings of the 41st International Conference on Software Engineering: Companion (ICSE-Companion)*, 2019, pp. 324–325.

[2] J. Bogatinovski, S. Nedelkoski, J. Cardoso, and O. Kao, "Artificial intelligence for IT operations (AIOps) on cloud platforms: Reviews, trends and open challenges," *IEEE Transactions on Cloud Computing*, vol. 11, no. 4, pp. 3607–3624, 2023.

[3] P. Notaro, J. Cardoso, and M. Gerndt, "A survey of AIOps methods for failure management," *ACM Transactions on Intelligent Systems and Technology*, vol. 12, no. 6, pp. 1–45, 2021.

[4] M. Du, F. Li, G. Zheng, and V. Srikumar, "DeepLog: Anomaly detection and diagnosis from system logs through deep learning," in *Proceedings of the 2017 ACM SIGSAC Conference on Computer and Communications Security (CCS)*, 2017, pp. 1285–1298.

[5] W. Meng, Y. Liu, Y. Zhu, S. Zhang, D. Pei, Y. Liu, and Y. Chen, "LogAnomaly: Unsupervised detection of sequential anomalies using deep learning," in *Proceedings of the 28th International Joint Conference on Artificial Intelligence (IJCAI)*, 2019, pp. 4727–4734.

[6] Y. Su, Y. Zhao, C. Niu, R. Liu, W. Sun, and D. Pei, "Robust anomaly detection for multivariate time series through stochastic recurrent neural network," in *Proceedings of the 25th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2019, pp. 2828–2837.

[7] H. Xu, W. Chen, N. Zhao, Z. Li, J. Bu, Z. Li, Y. Liu, Y. Zhao, and D. Pei, "Unsupervised anomaly detection via variational auto-encoder for seasonal KPIs in web applications," in *Proceedings of the 2018 World Wide Web Conference (WWW)*, 2018, pp. 187–196.

[8] A. Samir, C. Dehury, and P. Harsh, "Self-healing cloud microservices using advanced fuzzy logic," *IEEE Access*, vol. 7, pp. 156 542–156 555, 2019.

[9] J. Jiang, W. Lu, J. Chen, Q. Lin, P. Zhao, Y. Kang, H. Zhang, Y. Xiong, F. Gao, Z. Xu, Y. Dang, and D. Zhang, "How to mitigate the incident? an effective troubleshooting guide recommendation technique for online service systems," in *Proceedings of the 28th ACM Joint European Software Engineering Conference and Symposium on the Foundations of Software Engineering (ESEC/FSE)*, 2020, pp. 1410–1420.

[10] P. Wang, J. Xu, M. Ma, W. Lin, D. Pan, Y. Wang, and P. Chen, "CloudRanger: Root cause identification for cloud native systems," in *Proceedings of the 18th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID)*, 2018, pp. 492–502.

[11] J. Weng, J. Wang, J. Yang, and Y. Yang, "Root cause analysis of anomalies of multitier services in public clouds," in *Proceedings of the 26th International Symposium on Quality of Service (IWQoS)*, 2018, pp. 1–10.

[12] P. Chen, Y. Qi, P. Zheng, and D. Hou, "CauseInfer: Automatic and distributed performance diagnosis with hierarchical causality graph in large distributed systems," in *Proceedings of the 33rd IEEE International Conference on Computer Communications (INFOCOM)*, 2014, pp. 1887–1895.

[13] S. He, J. Zhu, P. He, and M. R. Lyu, "Experience report: System log analysis for anomaly detection," in *Proceedings of the 27th IEEE International Symposium on Software Reliability Engineering (ISSRE)*, 2016, pp. 207–218.

[14] Z. Li, Y. Zhao, J. Han, Y. Su, R. Jiao, X. Wen, and D. Pei, "InterFusion: Interacting hidden dynamics and macroscopic observations for multivariate time series anomaly detection," in *Proceedings of the 27th ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2021, pp. 3172–3180.

[15] S. Nedelkoski, J. Bogatinovski, A. Acker, J. Cardoso, and O. Kao, "Anomaly detection and classification using distributed tracing and deep learning," in *Proceedings of the 19th IEEE/ACM International Symposium on Cluster, Cloud and Grid Computing (CCGRID)*, 2019, pp. 241–250.

[16] G. Aceto, A. Botta, W. D. Donato, and A. Pescapè, "Cloud monitoring: A survey," *Computer Networks*, vol. 57, no. 9, pp. 2093–2115, 2013.

[17] M. T. Ribeiro, S. Singh, and C. Guestrin, "Why should I trust you? explaining the predictions of any classifier," in *Proceedings of the 22nd ACM SIGKDD International Conference on Knowledge Discovery and Data Mining (KDD)*, 2016, pp. 1135–1144.

[18] S. M. Lundberg and S. Lee, "A unified approach to interpreting model predictions," in *Advances in Neural Information Processing Systems 30 (NeurIPS)*, 2017, pp. 4765–4774.

[19] Z. Li, Y. Shu, Y. Gong, and Y. Zhao, "A survey on explainable anomaly detection," 2023.

[20] Y. Gan, Y. Zhang, K. Hu, D. Cheng, Y. He, M. Pancholi, and C. Delimitrou, "Seer: Leveraging big data to navigate the complexity of cloud-based debugging," in *Proceedings of the 24th International Conference on Architectural Support for Programming Languages and Operating Systems (ASPLOS)*, 2019, pp. 19–32.